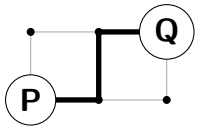
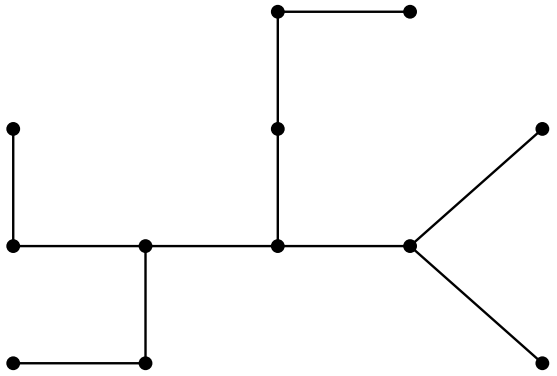
Each road between neighboring dots is one step. A shortest route uses the fewest steps. Roads may be drawn at different lengths.

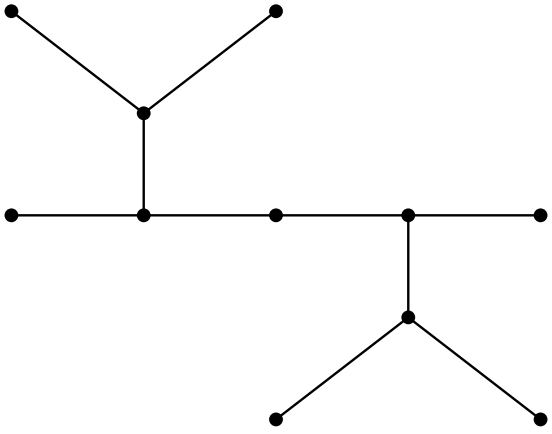


P to Q: 2 steps right, 1 step up.
This shortest route uses 3 roads.

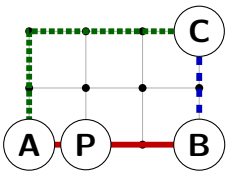
For three homes A, B, C, choose a shortest route for each pair. Color AB red, BC blue, and CA green. These three routes are a road triangle. Routes may overlap; draw the colors side by side when they do. Your partner checks that each route is shortest.

Problem 1: Choose three different dots for the homes on each tree. Draw the road triangles. Swap roles and try new homes. Can you make a road that belongs to exactly one of the three colored routes?



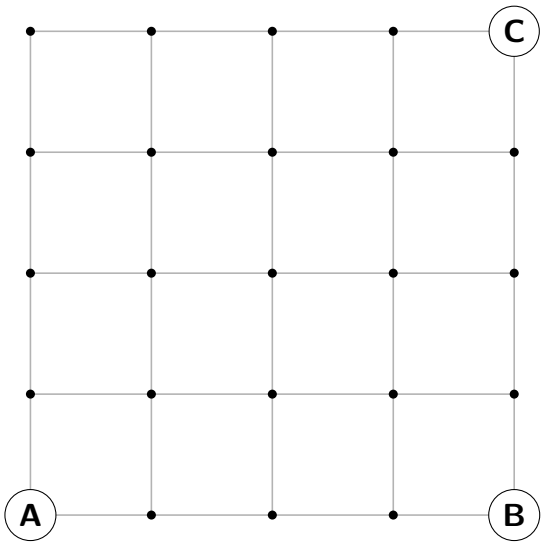


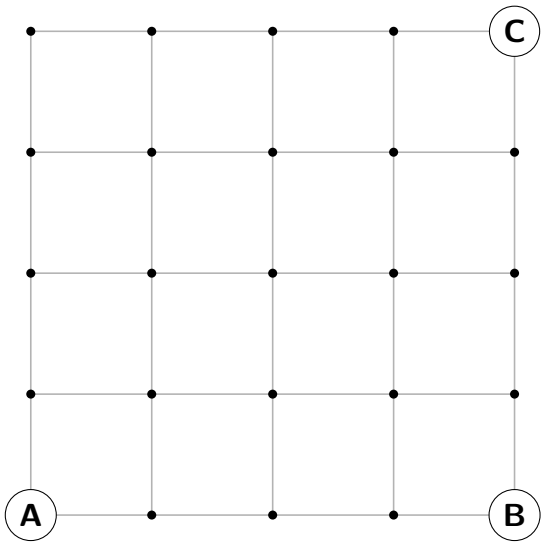
For a dot on one colored side, its **gap** is the fewest road steps needed to reach the nearer of the other two colored sides. Use any road on the map, including uncolored roads. A dot already on another side has gap 0.



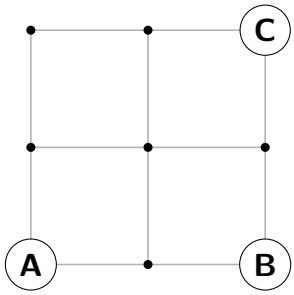
P is on the red side AB.
To green CA: 1 step left.
To blue BC: 2 steps right.
The smaller distance is 1: gap 1.

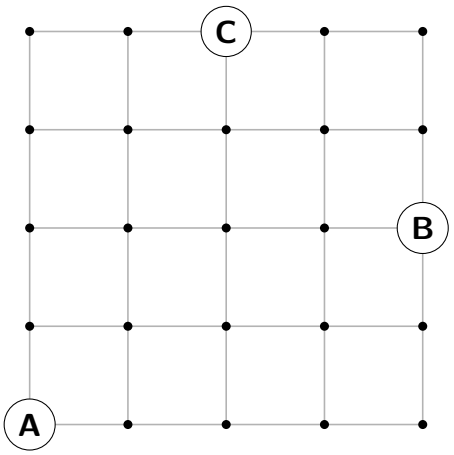
Problem 2: Choose shortest routes for the three sides on each map. Make every gap 0 on one map. On the other, make at least one gap 2 or more. Circle that dot and mark a shortest route to another side. Your partner checks the gap.

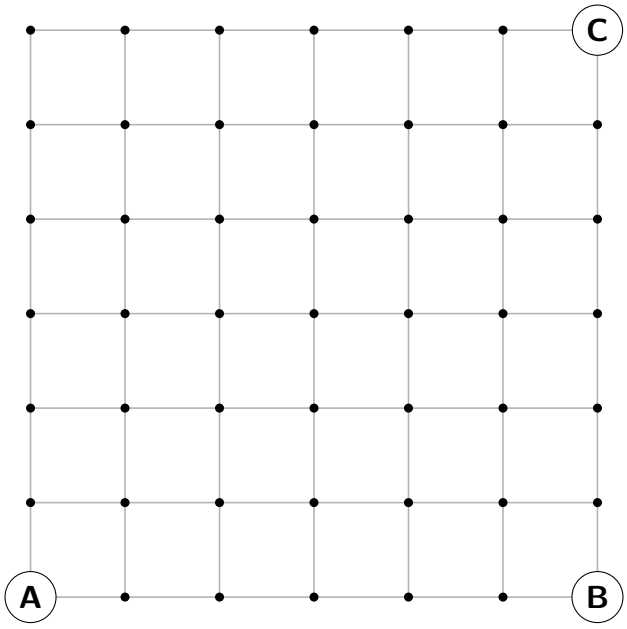




Problem 3: On each map, make the biggest gap as large as you can. All three colored sides must be shortest routes. Circle a dot with the biggest gap and write its gap beside it. Your partner tries to find a shorter route from that dot to another side.







Problem 4: Your partner names a whole number. Make a plan for a square-grid road triangle with a gap bigger than that number. You may use as large a grid as you need. Explain why all three sides are shortest and why your marked dot has the claimed gap.

Problem 5: A tree is a connected road map with no loops. Can any road triangle on any tree have a gap bigger than 0? Explain why your answer holds for every choice of three homes.